

A circular diagram representing the Machine Learning Life Cycle. It consists of eight interconnected nodes arranged in a circle, connected by a dark blue line. The nodes are: 'Business Need' (top, dark blue), 'Data' (top-right, dark blue), 'Develop' (right, dark blue), 'Train Model(s)' (bottom-right, dark blue), 'Test Model(s)' (bottom, dark blue), 'Deploy Model(s)' (bottom-left, dark blue), 'Monitor and Optimize' (top-left, dark blue), and 'Evaluate' (left, dark blue). The text 'Machine Learning Life Cycle' is written in a large, light blue font across the center of the diagram.

Beyond Kaggle: **Real-World** Data Science Consulting & Project Delivery

From Competition to Production: The Complete ML Project Lifecycle

The Reality Gap: Competition vs. Real-World ML

 **Kaggle**

 **Single metric** optimization (RMSE, AUC)

 **Leaderboard-driven** optimization

 **No deployment** required

 **No cost/latency** constraints

 **Little/no stakeholder** interaction

 **Real-World**

 **Multiple competing** KPIs & business constraints

 **Business value & risk** trade-offs

 **End-to-end deployment** essential

 **Resource optimization** critical

 **Frequent cross-team** collaboration

Executive Engagement: Translating Analytics to Business Value



Description

Industry ML succeeds when technical work is paired with **crisp storytelling** that secures sponsorship and adoption



Impact

Determines whether initiatives get **funded, launched**, and sustained in production



Ideation & Opportunity

Start with business pain points, validate data feasibility

- Frame proposals in executive KPI language



Benefit Crystallization

Translate technical gains into conservative financial cases

- Revenue lift, cost savings, risk mitigation



Executive Levers

Map work to scorecards of sponsors (CEO, CFO, COO, CMO)

- Highlight cross-functional wins



Problem Translation

Turn ambiguous asks into scoped briefs with clear decisions

- Document agreements to avoid scope creep



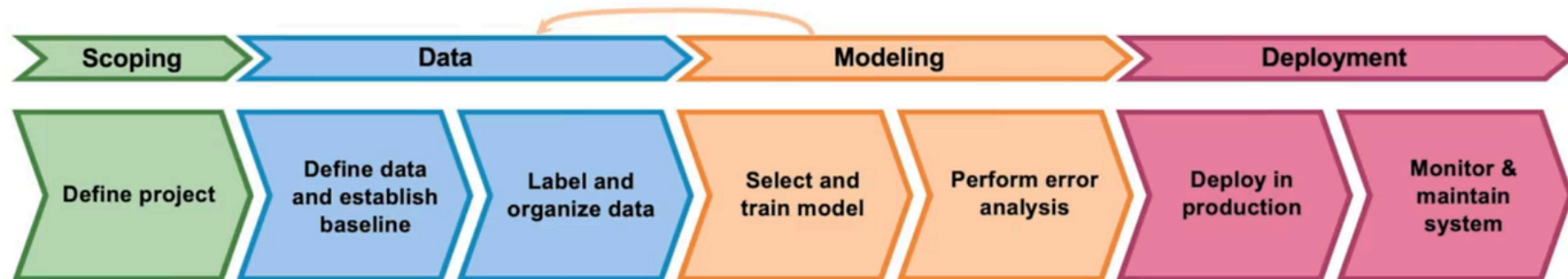
Presentation Craft

Lead with recommendation and quantified impact

- Business story: situation → complication → resolution

The ML Project Lifecycle

The ML project lifecycle



Problem Scoping

Translate business needs into **actionable ML tasks**



Data Engineering

Gather, clean, and **integrate data** from multiple sources



Model Specification

Define requirements, constraints, and **expectations**



Data Governance

Ensure **privacy**, security, and compliance



Deployment

Deliver models as **APIs**, **batch jobs**, or embedded systems



Monitoring

Track model drift, performance, and **system health**

*Real-world ML is an **iterative, collaborative process** that requires continuous refinement and adaptation*

Problem Scoping & Framing

Description

Translating **ambiguous business needs** into actionable ML tasks

Impact

Determines the **ceiling for achievable model performance**
Solving the wrong problem leads to wasted effort regardless of model quality

Effort Level

High - requires extensive stakeholder interviews and domain research

Effort: 

Modeling Implications

- Chosen framing affects data, features, and evaluation metrics
- Classification vs. regression decisions
- Time horizon and granularity considerations

Process

Iterative and ongoing — often revisited as new data or feedback emerges
Requires continuous refinement and stakeholder alignment

Data Acquisition & Engineering

Description

Gathering, cleaning, and integrating data from **multiple, often messy sources**

Impact

Data quality is the single biggest determinant of real-world model success

Improvements here often yield larger gains than model tweaks

Effort Level

High - often the **most time-consuming phase** of the project

Effort: 

Modeling Implications

- Enables richer feature sets
- Reduces noise and improves validation
- Poor pipelines lead to non-reproducible results

Process

End-to-end pipelines must be:

- ✓ **Automated** for consistency
- ✓ **Versioned** for reproducibility
- ✓ **Monitored** for changes

Model Specification: Overview

Description

Defining the **requirements, constraints, and expectations** for a machine learning solution

Impact

Critical for ensuring alignment with organizational goals, technical constraints, and regulatory requirements

Effort Level

Medium - requires careful coordination but typically less time-consuming than data engineering

Effort: 

Modeling Implications

Guides development of a **well-scoped model** that meets business objectives and technical constraints

Process

Creating a **detailed document** that outlines:

- Business objectives & KPIs
- Data sources & feature definitions
- Model architecture & evaluation metrics
- Deployment targets & resource constraints

Model Specification: Core Definitions

Business Objective & KPIs

Define the **business problem** and success metrics

- Quantifiable outcomes that drive value

💡 Example:

"Reduce customer churn by 15% in Q3"

Target Variable (y)

Precise definition of what you're **predicting**

- Clear calculation method and business meaning

💡 Example:

"Churn flag: 1 if customer cancels within 30 days"

Input Features (x)

Complete list of **predictors** with definitions

- Data sources, transformations, and availability

💡 Example:

"Tenure, usage metrics, support tickets count"

Time Periods

Training and testing data **timeframes**

Train

Historical data

Gap

Implementation lag

Predict

Future timeframe

💡 Example:

"Train: Jan-Jun, Gap: Jul, Predict: Aug-Sep"

Model Specification: Model Development

Data & Preprocessing

-  **Data sources** and access methods
-  **Preprocessing steps** for data quality
-  Feature **transformations** and engineering

Model Architecture

-  **Algorithm selection** and rationale
-  Hyperparameter **tuning strategy**
-  Model complexity and interpretability balance

Score Metric

- Primary metric for model **optimization**
-  Clear calculation method and business relevance
 -  Alignment with business objectives

Evaluation Metrics

Comprehensive **performance assessment**

-  Accuracy
-  Precision
-  Recall
-  F1 Score
-  MSE

 Thresholds for acceptable performance

Model Specification: Visualization & Communication

Visualization Types



Performance
Dashboards



Feature Importance
Plots



Prediction
Explanations



Data Drift
Visualizations



Business Metrics
Correlation



Statistical
Relationship
Analyses

Audience-Specific Views



Executive Summaries

High-level KPIs and business impact metrics for leadership



Technical Deep-dives

Model architecture, feature importance, and **performance trade-offs**



Operational Metrics

Practical dashboards with **actionable insights** for frontline teams



Data Exploration Reports

Visual representations of data patterns and **correlations**

Model Specification: Deployment & Operations

Deployment Targets

Define **where and how** the model will be deployed

 Batch

 Real-time

 Edge

 API

Resource Constraints

Specify **technical limitations** for deployment

 CPU/GPU requirements

 Memory and storage limits

 Latency and throughput targets



Monitoring & Retraining

Define **triggers and procedures** for model maintenance

 Performance degradation thresholds

 Data drift detection criteria

 Automated retraining schedule



Security & Reliability

Address **risk mitigation** and compliance requirements

 Rollback and failover procedures

 Privacy and data protection

 Regulatory compliance measures

Data Governance & Compliance

Description

Ensuring **data privacy, security**, and regulatory compliance

Global Standards

 GDPR  HIPAA  CCPA  PDPA

Effort Level

High - requires coordination with **multiple teams** and compliance with regulations

Effort: 

Impact

Non-compliance can halt projects or result in legal penalties
Regardless of model accuracy, compliance is non-negotiable

Modeling Implications

-  May **restrict feature use** and require anonymization
-  Limits data retention periods
-  May necessitate explainable models

Process

Ongoing with regular audits and documentation

-  Regular compliance reviews
-  Comprehensive documentation
-  Cross-functional collaboration

Model Deployment & Serving

Description

Delivering models as **APIs**, **batch jobs**, or embedded systems



API



Batch



Edge



Embedded

Impact

A model that scores well offline but **cannot be deployed reliably** delivers zero business value

Deployment quality directly affects user experience and operational success

Effort Level

High - can be **as much effort as model development**

Requires careful planning, testing, and infrastructure setup

Effort:

Modeling Implications

 May require **model optimization** for efficient formats

 Reducing latency for real-time applications

 Supporting rollback and versioning

Process

Includes **CI/CD**, **containerization**, and robust monitoring



Automated deployment pipelines



Containerization with Docker/Kubernetes



Comprehensive monitoring and alerting

Monitoring & Maintenance



Description

Tracking **model/data drift**, performance, and system health in production

↔ Data Drift

↘ Performance

🛡 System Health



Impact

Essential to catch silent failures before they impact business outcomes

Prevents model degradation and maintains business value



Effort Level

Medium - requires **initial setup** but becomes more automated over time

Effort:



Modeling Implications



May require building **drift detectors**



Alerting systems for anomalies



Fallback mechanisms for failures



Process

Continuous with regular retraining and validation



Performance metrics tracking



Data distribution monitoring



Scheduled model validation

Retraining Pipelines

Description

Automating **retraining, validation**, and deployment of updated models

 Scheduled

 Trigger-based

 Continuous

Impact

Enables **continuous improvement** and adaptation to changing data

Prevents model degradation and maintains business value over time

Effort Level

High initial setup; moderate ongoing maintenance

Effort: 

Modeling Implications

 Must ensure **reproducibility** of training process

 Versioning for model tracking

 Safe rollback mechanisms

Process

End-to-end automation, often orchestrated via workflow tools

 Airflow

 Kubeflow

 MLflow

 Automated validation before deployment

Stakeholder Communication

Description

Explaining **results, risks, and limitations** to non-technical teams

 Executives

 Business Teams

 Compliance

Impact

Essential for project buy-in, adoption, and alignment with business goals

Clear communication bridges the gap between technical work and business value

Effort Level

Continuous - requires regular updates and presentations to various stakeholders

Effort:

Modeling Implications

 May drive the choice of **interpretable models**

 Requires effective visualizations

 Focus on business-relevant metrics

Process

Ongoing with presentations, reports, and dashboards

 Regular progress updates

 Results interpretation in business terms

 Risk and limitation transparency

Cost & Latency Constraints



Description

Optimizing for **inference speed**, **memory**, and hardware costs



Latency



Memory



Cost



Impact

Directly affects **user experience** and operational expenses
Sometimes, a "worse" model is chosen because it is faster or cheaper



Effort Level

Variable - depends on specific performance requirements and constraints

Effort:



Modeling Implications



May need **model quantization** for efficiency



Distillation techniques for smaller models



Pruning to reduce complexity



Process

Often **iterative**, with trade-offs between accuracy and efficiency



Balancing model performance with constraints



Performance testing under realistic conditions



Hardware-specific optimizations

Key Takeaways



Real-world ML is a **holistic, collaborative process** that extends far beyond model development



Business Alignment

Success depends on solving the **right problem** with clear business objectives



Data Quality

Data engineering often yields **greater impact** than algorithmic improvements



Robust Processes

Comprehensive specifications and **automated pipelines** ensure reliability



Continuous Improvement

Monitoring, retraining, and **experimentation** maintain long-term value

"The best model is the one that delivers real business value in production"

Scenarios: From Problem to ML Plan

Advanced business scenarios involving modern machine learning algorithms and complex business problems



Retail Forecasting



Cybersecurity



Customer Segmentation



Ad-Tech CTR



Genomics



Supply Chain

Scenario 1: Retail/CPG - Sales Forecasting

Gradient Boosting for Weekly Sales Prediction

Business Objective

Forecast sales for **thousands of products** across hundreds of stores on a weekly basis
↗ 5% accuracy improvement = millions in savings

 Inventory optimization

Target Variable

Regression problem predicting weekly sales quantity
 Unit: Number of units sold per week for each SKU-store pair
 12-week forward forecast, updated weekly

Data Sources

3+ years of historical data
 POS transactions, product/store master data
 Promotional calendar, weather, local events

Model Selection

LightGBM Regressor
✓ Handles mixed feature types efficiently
✓ Fast training enables frequent retraining
✓ Captures complex non-linear relationships

Feature Engineering

Lag features for recent sales patterns
 Temporal features (week, month, season)
 Product/store attributes and hierarchies
 Promotion and external factors

Implementation

Weekly batch prediction pipeline
 Primary metrics: MAPE, WMAPE, RMSE
 Interactive forecast visualization dashboard
 Automated retraining with rolling window

Scenarios 2-6: Learner Exercises



S2. Cybersecurity: Network Intrusion Detection

A financial services company needs to protect their digital infrastructure by detecting **malicious network activity in real-time**, with particular emphasis on identifying novel and previously unseen attacks (zero-day exploits). This solution will enhance the company's security posture, reduce risk of data breaches, maintain regulatory compliance, and protect customer assets.

↗ Early detection can save millions in potential breach costs



S3. Customer Analytics: High-Dimensional Segmentation

A large B2C company aims to develop a deep understanding of their customer base by identifying **distinct customer personas** from high-dimensional behavioral data. These customer segments will enable personalized user experiences, targeted marketing campaigns, product recommendations, and strategic business planning.

↗ Drive increased conversion rates and customer lifetime value



S4. Ad-Tech: Predicting Ad Click-Through Rate

An ad-tech company aims to predict the probability that a user will click on an ad in **real-time**. This prediction powers real-time bidding systems to optimize ad selection and bid amounts, maximizing ROI for advertisers while improving user experience through more relevant ads.

↗ 0.1% accuracy improvement = millions in additional revenue



S5. Genomics: Cancer Subtype Classification

A biomedical research institution wants to classify different subtypes of cancer based on **gene expression data**. Accurate classification is essential for personalized medicine, where treatment is tailored to the specific cancer subtype. The solution will support pathologists in diagnosing cancer subtypes more accurately and efficiently.

↗ Potentially leading to better patient outcomes and treatment plans



S6. Supply Chain: Demand Forecasting

A CPG company wants to improve its demand forecasting for a **popular beverage** to optimize inventory and reduce stockouts. The forecast should account for seasonality, holidays, promotions, and external factors like weather. This will help balance inventory levels to meet demand while minimizing waste and storage costs.

↗ Optimize inventory levels and reduce waste across the supply chain



Exercise Framework

For each scenario, develop a complete **ML plan** that addresses the business problem. Your plan should include:

- ✓ Clear problem definition and success metrics
- ✓ Data requirements and feature engineering approach
- ✓ Model selection and validation strategy
- ✓ Implementation plan and visualization approach



These exercises help you apply the concepts from the presentation to real-world ML problems